FORENSIQ RESEARCHER ACTION GUIDE

Step-by-Step Protocols to Fulfill All Audit Directives from ForensIQ (2).pdf

Audience: Lead Forensic Researcher, Evaluator, and Paper Author
Target Case: NIST CFReDS Data Leakage Scenario (Iaman Informant / OOO Corp)
Companion Codebase: ForensIQ Canonical Engine (12-Stage Pipeline + Baseline Benchmark)
Purpose: Concrete instructions detailing exactly what the human researcher must execute, verify in third-party tools, document in research logs, and write in the publication paper.

1. Executive Summary: What Changed & Core Scientific Truths

The comprehensive audit (ForensIQ (2).pdf) identified that the original project relied on circular ground-truth assertions (author-written regexes plus a hardcoded timestamp positive) to report an unvalidated **71.43% recall** figure. Furthermore, a simple rule selecting .exe files under /Program Files found 17 documented executables after inspecting just 76 files, challenging whether unsupervised machine learning added measurable value at low review budgets.

All technical bugs in the codebase have now been corrected:
• Unallocated deletion bitmask fixed from equality to bitwise AND.
• Missing timestamp zeros are masked to prevent artificial ~34-year timestomping anomalies.
• Fitted StandardScaler is permanently serialized alongside the model.
• Module-level execution on import has been eliminated across all extractors.
• A scientific baseline benchmark (evaluate_baselines.py) has been built as Pipeline Stage 12.

2. Mandatory Action Checklist for the Researcher

To complete your defense, thesis, or paper publication, complete each of the following 6 tasks sequentially:

Task ID	Action Required by Researcher	Target Artifact / Tool	Verification / Deliverable
TASK 1	**Verify Source Image Provenance:** Record official SHA-256 and MD5 hashes from the NIST CFReDS portal before mounting or hashing.	NIST CFReDS portal / cfreds_2015_data_leakage_pc.raw	Provenance table in paper with official reference hashes.
TASK 2	**Run the 12-Stage Pipeline:** Execute the automated pipeline on the suspect image and ensure all outputs generate without errors.	py -3.12 run_pipeline.py --start-at 3	Verify exit code 0; inspect reports/forensiq_report.html.
TASK 3	**Execute Milestone 5 Benchmark:** Run equal-budget comparisons against Random, Rules, Univariate, and k-NN baselines.	py -3.12 evaluate_baselines.py	Generate reports/baseline_budget_comparison.png and Markdown table.
TASK 4	**Manual Examiner Triage (FTK/Autopsy):** Verify top 25 anomalous files in FTK Imager or Autopsy to validate actual evidentiary value.	FTK Imager / Autopsy / Hex Editor	Record confirmation log of true positives vs OS infrastructure.
TASK 5	**Security Log Gap Investigation:** Inspect why Security.evtx has 0 events during work hours (disabled auditing vs cleared log).	Event Viewer / artifacts/evtx/	Check for Event ID 1102 (cleared log) vs disabled audit policy.
TASK 6	**Write Paper with Honest Claims:** Remove all mentions of '71.43% recall'. Frame IF as multi-modal triage discovering 7 targets at K=4250.	LaTeX / Paper Manuscript	Include equal-budget table and ATT&CK; Evidence Cards.

3. Cross-Image Partition Offsets & Execution Matrix

The PDF audit requires multi-image cross-validation. When evaluating different forensic disk images, ensure the correct sector offset is configured in config.py:

Dataset / Forensic Image	OS Version	Partition Sector Offset	Filesystem & Log Format	Active Engine
`NIST CFReDS Data Leakage PC`	Windows 7 x64 SP1	**`206848 (Partition 2)`**	NTFS / Modern .evtx	extractor_for_evtx.py
`NIST CFReDS SCHARDT Case`	Windows XP SP2	**`63 (Partition 1)`**	NTFS / Legacy .evt	extractor_for_evt.py
`Lone Wolf Scenario (Image 3)`	Windows 10 x64	**`2048 or Auto`**	NTFS / Modern .evtx	extractor.py (GPT Auto)

4. Scientific Benchmark Interpretation (Milestone 5 Results)

The empirical equal-budget benchmark executed on the 85,017 files of the NIST CFReDS dataset produced the following validated results across 7 investigation budgets:

Ranking Method	K=25	K=50	K=100	K=250	K=500	K=1000	K=4250 (5%)
Random Baseline (10 seeds)	0.0	0.0	0.0	0.0	0.0	0.1	1.3±1.2
Rule: .exe under /Program Files	0/19	0/19	0/19	0/19	0/19	1/19	1/19
`Rule: Suspicious Paths (User/Temp)`	0/19	0/19	1/19	**`12/19`**	**`12/19`**	**`12/19`**	**`13/19`**
Univariate: delta_mc	0/19	0/19	0/19	0/19	0/19	0/19	0/19
Univariate: File Size	0/19	0/19	0/19	0/19	0/19	1/19	6/19
`Isolation Forest (17 Feats)`	0/19	0/19	0/19	0/19	0/19	0/19	**`7/19`**
Isolation Forest Multi-Seed (0–9)	0.0	0.0	0.0	0.0	0.0	0.0	6.6±1.6
`k-NN Distance Outlier (k=5)`	0/19	0/19	0/19	0/19	3/19	4/19	**`12/19`**

Key Takeaways for Your Research Paper:
*1. **Why Rules Win at Low K (K <= 250):** Domain rules directly encode examiner experience (focusing on /Users/), while unsupervised density estimators rank giant Windows registry hives and database files as the most extreme mathematical anomalies.*
*2. **Where ML Adds Genuine Value (K = 4250 / 5% Triage):** Narrow rules (e.g. .exe under /Program Files) completely miss 16 of the 19 targets because modern insider threats involve non-executable files (databases, LNK shortcuts, and exfiltrated docs). Isolation Forest successfully surfaces 7 targets, and k-NN surfaces 12 targets across multi-modal indicators.*

5. Guidelines for Writing the Research Paper Narrative

When authoring the manuscript for conference or journal submission, adhere strictly to these principles:
• **No Binary Negative Claims:** Prominently state in the Methodology: *'Because raw forensic disk images lack independently verified negative labels for every operating system file, the task is framed as an unsupervised investigation prioritization problem, not binary classification.'*
• **Equal-Budget Evaluation:** Report Precision@K and Recall@K at matched review budgets (K=25, 50, 100, 250, 500, 4250). Avoid reporting only single-contamination metrics.
• **Framing ATT&CK; Assertions:** Do not claim 'confirmed' timestomping or USB exfiltration from single MFT attributes or registry keys alone. Present them as *hypotheses requiring secondary corroboration* (e.g. comparing $SI vs $FN attributes or checking Shellbags).
• **Evidence Integrity:** Accurately describe chain_of_custody.py as an *automated cryptographic manifest verification system*, not an ISO/IEC 27037 legal chain of custody.

6. Critical Safety Protocols: What NOT to Do

WARNING — Section 9.1 Directive:

*1. **DO NOT execute remediation PowerShell scripts** (reports/remediation/remediate_*.ps1) on the evidence workstation or any live forensic image. They modify system registries, kill accounts, and destroy evidentiary timestamps.*

*2. **DO NOT modify raw forensic disk images.** Always verify images in read-only mode.*

*3. **DO NOT report 71.43% recall** in any presentation, report, or slide deck.*